Same Message, Different Reading? Testing Whether LLM-Labelled Political Opinion Travels Across Scripts and Platforms in Pakistan and India

Research proposal, PhD in Communications and New Media, National University of Singapore (August 2027 intake) · Asad Ikram · asad.ikram53@gmail.com
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1. Problem

Large language models are becoming the default instrument for reading political opinion from social media (Gilardi, Alizadeh and Kubli, 2023). We already know that this instrument is sensitive: the same labels change with the model and prompt, often enough to reverse conclusions (Baumann et al., 2025), and its errors can be decomposed into sources such as prompt and judge model (Messing, 2026). We also know that multilingual text analysis must show equivalence across languages (Lind et al., 2025), and that the platform itself is a named source of error in digital trace data (Sen et al., 2021). Evidence is missing where these problems meet. In Pakistan and India the same political talk appears in Urdu script, Roman Urdu, English and Hinglish, and different audiences comment under the same news story on YouTube and Facebook. If an LLM reads a Roman Urdu comment differently from the same comment in Urdu script, or reads a YouTube comment section differently from a Facebook one, comparisons may reflect the instrument rather than the public.

2. Prior work and my starting point

Two lines of work frame the project. Communication research shows that platform differences can be real: commenters self-select, and more uncivil users comment more (Kim, Guess, Nyhan and Reifler, 2021), while work at CNM shows that the relation between engagement metrics and news production differs by platform (Mukerjee, Yang and Peng, 2023; Ouyang, Mukerjee and Yang, 2026). Measurement research shows that labels can shift by language on identical content, which is only visible with designs that hold the content fixed (Nogara et al., 2025), and that romanised Indian-language text can be handled worse than native script (Khullar et al., 2025), although other studies find LLM political labels similar across languages (Hartman and Törnberg, 2026). The core problem is identification: a gap between platforms or scripts may be a real difference in what people say or an error in how the model reads them. Correction methods with a human sample (Egami et al., 2023) help only when the human reference is valid for each group.

With Dr Ella Haig I recently released a protocol for testing whether the platform acts as a source of measurement error, by analogy to survey modes (Ikram and Haig, 2026, arXiv preprint). Its known-truth simulation shows how pooling misleads when platforms measure differently and the platform mix shifts, and why platform differences can be identified only relative to each other. It has not been applied to real data. It grew out of my MSc dissertation (255,446 comments across four platforms), where my classifier's 88 per cent "accuracy" turned out to be agreement with the lexicon labels it was trained on; no human-validated accuracy existed.

3. Research questions

1. **Scripts.** When the same political message is written in Urdu script, Roman Urdu, English and Hinglish, do LLM annotators assign the same stance and sentiment as Pakistani and Indian coders, and does the answer depend on the model?
2. **Platforms.** When the same outlet or party posts the same story on YouTube and Facebook, how much of the gap in comment stance is LLM error that differs by platform, and how much is a difference in what audiences express?
3. **Consequences.** When labels are corrected for the non-equivalence found in RQ1 and RQ2, does a substantive Pakistan-India comparison around fixed events change, and at what sample sizes would it?

4. Data and methods

Study 1 (year 1, no platform access needed). With native speakers, build about 300 political messages in four parallel versions (Urdu script, Roman Urdu, English, Hinglish), plus naturally occurring bilingual pairs where politicians post the same message in two languages. Trained coders label stance and sentiment: Pakistani coders

for Urdu and Roman Urdu, Indian coders for Hinglish, both for English. Three or four frozen open-weight LLMs and one fine-tuned code-mixed model label the same items. A rater model that treats coders and models as raters (Dawid and Skene, 1979) estimates whether model-human disagreement differs by script, and a design study sets how many coders, items and models the later studies need. This produces a Roman Urdu and Urdu political-stance gold set, extending Hinglish resources such as Swami et al. (2018) and SemEval-2020 SentiMix (Patwa et al., 2020).

Study 2 (years 2 to 3). Collect comments under the same stories posted by 10 to 20 matched news outlets and party accounts per country on YouTube (YouTube Researcher Program) and Facebook Pages (Meta Content Library, with the same open-weight models run inside Meta's secure environment), with a small paid X sample. Collection is prospective, around recurring events fixed in advance such as Pakistan's federal budget and India's Union Budget. A stratified sample per platform and script is double-coded. Because the story is held fixed, LLM error that differs by platform can be measured against human labels and separated from the remaining audience difference.

Study 3 (years 3 to 4). Re-run a Pakistan-India comparison of reaction to these events with uncorrected and corrected labels, and report when the conclusion changes and when it does not. A finding of no meaningful difference is a result, because it would support comparisons the field already makes. Ethics: NUS ethics review, public pages only, no user identities, paraphrased quotes.

5. Why NUS CNM, and feasibility

The project sits in computational and political communication, across CNM's Politics, Participation and Governance and Technology, Design and Social Futures clusters. Dr Subhayan Mukerjee's work on news, audiences and platforms and on India make him the natural primary supervisor; RQ2 moves his platform-dependent findings from news production to the measurement of audience reaction. Dr Zening Duan's work on how measures are operationalised (Duan et al., 2025) and on comparison across platforms (Chen, Duan and Kim, 2024) connects to RQ1. Work by Associate Professors Taberez Neyazi and Kokil Jaidka on South Asian political communication, including the call by Jaidka et al. (2019) for Urdu and Hindi resources, and by Assistant Professor Ozan Kuru on public opinion measurement, would inform RQ3.

On feasibility: I have collected and labelled public web and social data for nearly eight years, at Dubizzle Labs (data in English, Arabic, Urdu and Hindi across nine countries), at M+C Saatchi Fluency (LLM tagging with prompt-version tracking) and as co-founder and CTO of Artemis AI, where our own evaluation ran into a problem now documented at scale: agreement between models is not accuracy (Kim et al., 2025). I speak Urdu and Hindi, which helps with codebook and gold-set design; coding will be done by trained Pakistani and Indian coders.

6. Expected contribution

The contribution is applied rather than a new method: a test of whether LLM opinion labels mean the same across South Asian scripts and platforms, using designs that hold the message fixed; a Roman Urdu and Urdu political-stance gold set; and evidence on whether these differences change substantive conclusions, including where they do not. A first paper is planned for *Computational Communication Research*.

Selected references

Baumann, J. et al. (2025). Large language model hacking: quantifying the hidden risks of using LLMs for text annotation. arXiv:2509.08825.

Chen, K., Duan, Z. and Kim, S. J. (2024). Uncovering gender stereotypes in controversial science discourse: evidence from computational text and visual analyses across digital platforms. Journal of Computer-Mediated Communication 29(1), zmad052.

Dawid, A. P. and Skene, A. M. (1979). Maximum likelihood estimation of observer error-rates using the EM algorithm. Journal of the Royal Statistical Society C 28(1).

Duan, Z. et al. (2025). Constructing vec-tionaries to extract message features from texts: a case study of moral content. Political Analysis.

Egami, N., Hinck, M., Stewart, B. M. and Wei, H. (2023). Using imperfect surrogates for downstream inference: design-based supervised learning for social science applications of large language models. NeurIPS.

Gilardi, F., Alizadeh, M. and Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. PNAS 120(30).

Hartman, V. and Törnberg, P. (2026). Measuring negative campaigning across languages with large language models: a study of 18 million tweets in 19 countries. arXiv:2507.17636.

Ikram, A. and Haig, E. (2026). Is the platform part of the measurement? A protocol and simulation study for cross-platform equivalence of machine-labelled policy sentiment. arXiv:2610.16757.

Jaidka, K., Ahmed, S., Skoric, M. and Hilbert, M. (2019). Predicting elections from social media: a three-country, three-method comparative study. Asian Journal of Communication 29(3).

Khullar, M. et al. (2025). Script gap: evaluating LLM triage on Indian languages in native vs romanized scripts in a real world setting. arXiv:2512.10780.

Kim, E. M., Garg, A., Peng, K. and Garg, N. (2025). Correlated errors in large language models. ICML, PMLR 267.

Kim, J. W., Guess, A., Nyhan, B. and Reifler, J. (2021). The distorting prism of social media: how self-selection and exposure to incivility fuel online comment toxicity. Journal of Communication 71(6).

Lind, F. et al. (2025). Grounding the comparative turn in communications: a framework for validating multilingual computational text analysis. Computational Communication Research 7(1).

Messing, S. (2026). Hidden measurement error in LLM pipelines distorts annotation, evaluation, and benchmarking. arXiv:2604.11581.

Mukerjee, S., Yang, T. and Peng, Y. (2023). Metrics in action: how social media metrics shape news production on Facebook. Journal of Communication 73(3).

Nogara, G. et al. (2025). Toxic bias: Perspective API misreads German as more toxic. ICWSM.

Ouyang, R., Mukerjee, S. and Yang, T. (2026). How social media metrics shape news production: a replication study. The International Journal of Press/Politics.

Patwa, P. et al. (2020). SemEval-2020 Task 9: overview of sentiment analysis of code-mixed tweets. SemEval.

Sen, I., Flöck, F., Weller, K., Weiß, B. and Wagner, C. (2021). A total error framework for digital traces of human behavior on online platforms. Public Opinion Quarterly 85(S1).

Swami, S. et al. (2018). An English-Hindi code-mixed corpus: stance annotation and baseline system. CICLing.